

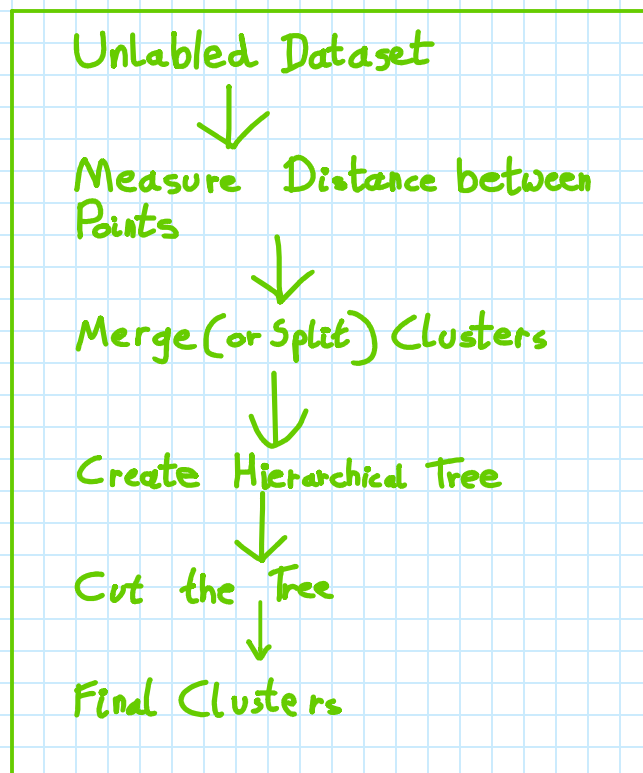
Hierarchical Clustering

1. What is Hierarchical Clustering?

Hierarchical Clustering is an **Unsupervised Machine Learning Algorithm** that groups similar data points into clusters by creating a **hierarchical tree-like structure**, called a **Dendrogram**.

Unlike K-Means, Hierarchical Clustering **does not require the number of clusters (K) before training**. Instead, it builds a hierarchy of clusters, and you decide the final number of clusters by cutting the dendrogram.

Main Idea



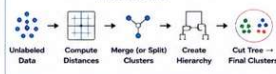
HIERARCHICAL CLUSTERING

An Unsupervised Learning Algorithm that Builds a Hierarchy of Clusters (Tree)

1. WHAT IS HIERARCHICAL CLUSTERING?

- Hierarchical Clustering creates a hierarchy of clusters by either merging or splitting them.
- It does not require the number of clusters (K) before training.
- The result is a tree-like diagram called a Dendrogram.
- We decide the final number of clusters by cutting the dendrogram at a suitable height.

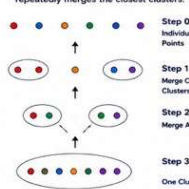
HOW IT WORKS



2. TYPES OF HIERARCHICAL CLUSTERING

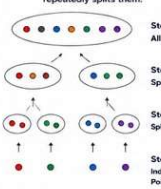
A. AGGLOMERATIVE (BOTTOM-UP)

Starts with each point as a single cluster and repeatedly merges the closest clusters.



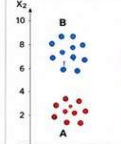
B. DIVISIVE (TOP-DOWN)

Starts with all points in one cluster and repeatedly splits them.

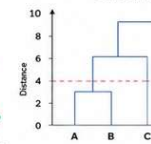


3. DENDROGRAM EXAMPLE

Sample Data (2D)



Dendrogram



- How to Decide Number of Clusters?
- Draw a horizontal line (red dashed line).
 - Count the number of vertical lines (branches) that the line cuts.
 - Here it cuts 2 branches $\rightarrow K = 2$ clusters

4. LINKAGE CRITERIA (How to Measure Distance between Clusters)

A. SINGLE LINKAGE (Nearest Neighbor)

Distance between the closest pair of points in the two clusters.

$$D(A,B) = \min_{a \in A, b \in B} d(a,b)$$

Good for: Irregular shapes
Drawback: Chaining effect

B. COMPLETE LINKAGE (Farthest Neighbor)

Distance between the farthest pair of points in the two clusters.

$$D(A,B) = \max_{a \in A, b \in B} d(a,b)$$

Good for: Compact clusters
Drawback: Sensitive to outliers

C. AVERAGE LINKAGE (UPGMA)

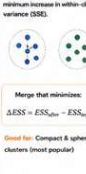
Average distance between all pairs of points in the two clusters.

$$D(A,B) = \frac{1}{|A||B|} \sum_{a \in A} \sum_{b \in B} d(a,b)$$

Good for: Balanced clusters

D. WARD'S LINKAGE (Minimum Variance)

Merges clusters that result in the minimum increase in within-cluster variance (SS).



5. STEP-BY-STEP WORKING (Agglomerative Example)

Example Data (2D)

Point ID	X	Y
A	1	2
B	2	1
C	8	7
D	7	8

Step 0: Each point is a cluster

Step 1: Merge Closest (A,B)

Step 2: Compute Distance Matrix

	A	B	C	D
A	0	1.41	8.60	9.22
B	1.41	0	8.06	8.49
C	8.60	8.06	0	1.41
D	9.22	8.49	1.41	0

Step 3: Recompute Distances

	A,B	C	D
A,B	0	8.19	8.73
C	8.19	0	1.41
D	8.73	1.41	0

Step 4: Merge Closest (C,D)

Step 5: Merge Final

Dendrogram represents all these merge steps.

6. IMPLEMENTATION (Python - scikit-learn)

```
from sklearn.cluster import AgglomerativeClustering
from sklearn.preprocessing import StandardScaler
from scipy.cluster.hierarchy import dendrogram, linkage
import matplotlib.pyplot as plt

# 1. Data
X = [[1,2], [2,1], [8,7], [7,8]]

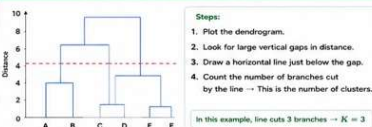
# 2. Scaling
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# 3. Agglomerative Clustering
model = AgglomerativeClustering(n_clusters=2, # final clusters
                                metric='euclidean',
                                linkage='ward')

labels = model.fit_predict(X_scaled)
print('Cluster Labels:', labels)

# 4. Dendrogram
linked = linkage(X_scaled, method='ward', metric='euclidean')
plt.figure(figsize=(7,4))
dendrogram(linked, labels=[ 'A', 'B', 'C', 'D' ])
plt.title('Dendrogram (Hierarchical Clustering)')
plt.xlabel('Data Points')
plt.ylabel('Distance')
plt.show()
```

7. CHOOSING NUMBER OF CLUSTERS



8. SUMMARY

- Builds a hierarchy of clusters (tree structure).
- Does not require K in advance.
- Agglomerative (bottom-up) is most commonly used.
- Uses distance metrics and linkage criteria to merge clusters.
- Dendrogram helps to understand cluster structure and choose K.
- Sensitive to outliers and computationally expensive for large datasets.



9. ADVANTAGES

- ✓ No need to specify K before training
- ✓ Dendrogram provides rich information
- ✓ Works well for small datasets
- ✓ Can handle any shape of clusters (with appropriate linkage)

10. DISADVANTAGES

- ✗ Computationally expensive ($O(n^3)$)
- ✗ Sensitive to noise and outliers
- ✗ Hard to undo merges (once merged)
- ✗ Not suitable for very large datasets